# L4 Safety Island for Autoware

## Software design, Part 1 — node allocation

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THE AUTOWARE FOUNDATION

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# Revision history

Major revisions only. The version is a date-code, YYYYMMDD, and all four artefacts of this deliverable — design document, this deck, the review edition, the discussion document — carry the same one.

| Version | Issued | Status and scope |
|---|---|---|
| **20261023** | 2026-10-23 | **Major revision**, from the WG session of 2026-10-17. **Island sensors ranked**: camera mandatory, then radar, ultrasonic, IMU / GNSS, 2D lidar optional; B4 is the loss of the camera. B2 first slows to the **ODD's safe speed** (~60 km/h highway, ~30 km/h city), holds **8–15 s**, then B3 if the camera still sees the lane, else B4. B3 finds its space with the camera. B5 tele-operation and B6 joystick control declared on a new RCC link and local control port, with a stage-1 detailed design. **Two design branches**, Least-Cost and Maximum-Safety, with every behaviour adjusted to each. Open decisions 19–23 added |
| **20260918** | 2026-09-18 | **Major revision.** B1 recast as an **ODD switch negotiation** — the island declares the current domain unsatisfiable, the HPC picks one it can meet, the island verifies it; B1 gains both exits and the safety timeout $T_{\text{odd}}$. $v_{\text{hold}}$ defined as a **decaying ramp** rather than a constant. B3 gains an **island-detected refuge** when H8 offers none, bounded by $T_{\text{refuge}}$. Open decisions 17 and 18 added |
| 20260912 | 2026-09-12 | **Editorial revision.** §4 of the design gains a five-phase figure series elaborating the node allocation. No normative change |
| 20260903 | 2026-09-03 | **Major revision.** B4 declared as the fourth behaviour; the optional HPC object list admitted; B3 and B4 both recognised as terminal; hazard lamps raised at entry; the abort ladder drawn; open decisions 15 and 16 added |
| 20260827 | distributed 2026-08-28 | First WG review draft. Three declared behaviours B1–B3, the node allocation, the interface contract, and fourteen open decisions |

**Takeaway.** Everything new since 20260827 is marked in marker yellow. A reviewer of that set can read the marking alone and see the whole change.

# Agenda

1. Where we are
2. What the island must do
3. How the allocation was decided
4. The allocation
5. Island sensing (D7)
6. Pull-over MRM
7. ODD degradation
8. Interfaces and deployment
9. Minimal configuration
10. Two design branches
11. Decisions and roadmap

# Reading the identifiers

The prefix says what *kind* of thing an identifier is — and what it commits the design to.

| Symbol | Names | The property it carries | In this deck |
|---|---|---|---|
| **D** *n* | WG *decision* | **Settled before Part 1 began.** An *input* to the allocation, not something this deck may reopen | D1–D7 |
| **B** *n* | Declared *behaviour* | **Normative, and the only normative thing here.** What the island does, seen from outside. Every allocation below must trace to one | B1–B4 |
| **R** *n* | Partitioning *rule* | **A test, applied per node** — which side of the link it runs on. R5 decides most of them | R1–R8 |
| **T** *n* | Criticality *tier* | **An integrity class and a dependency constraint:** the ASIL a node is built to, and — the part that bites — what it may depend on | T0 T1 T1s T2 T3 |
| **P** *n* | Execution *partition* | **A scheduling unit, not an integrity class.** One executor at a fixed priority band; the numbering is the order the island *sheds* them under overload | P0–P4 |
| **I** *n* | *Imported* node | **Exists upstream today.** Migrated to the island; the argument for it is *provenance*, so it moves unmodified or not at all | 18 on the target path |
| **N** *n* | *New* island node | **Has to be written.** Named `si_*`; nothing upstream provides it, so each is a cost to justify against a behaviour | 19 new |
| **H** *n* | New *HPC* node | **Has to be written — by the other team.** What the island needs the HPC to publish: where this design asks something of someone not in the room | 9 new |

**A chip's colour is its tier**

`T0` core · `T1` supervision · `T1s` sensing · `T2` nominal · `T3` non-real-time · `HPC` other domain. The chip says the tier, not the prefix: `I6` and `I10` are imported packages at different integrity, `P2` a *partition* at T1s. **T is how well a node is built; P is when it runs.** Every chip is a link.

**D1 is not "decision 1"**

Two unrelated sequences share the letter. **D** *n*, no hyphen, is a WG input and is *closed*. **Open decision** *n* — **D-*nn*** in the discussion document — is still to be settled; those are the last three slides.

Where we are

What the WG has already decided, and what is already running on the bench.



# Inputs already fixed by the WG

| # | Decision |
|---|---|
| D1 | The L4 Safety Island is a **standalone, safety-certified ECU** — not an on-SoC partition |
| D2 | HPC hosts perception / planning / VLM; the island hosts **status monitoring, the control module, and MRM** |
| D3 | HPC ↔ island is **TSN / Vehicle Ethernet**; island → actuators is CAN FD / FlexRay |
| D4 | Island middleware is **nano-ros** — zenoh-pico core, ROS-message compatible, RTOS-portable |
| D5 | Island OS and nodes must be **real-time compliant**: predictable execution time, memory safe |
| D6 | Every safety-relevant payload carries **end-to-end protection** — CRC, counter, data ID, age; gPTP provides the time base. **AUTOSAR is out of scope: Autoware in-house design** |
| D7 | **Low-resolution, high-robustness sensors connect directly to the island** — 2D lidar, forward camera, ultrasonic, radar — so the MRM nodes execute without the HPC |
| D8 | **Not every island carries every sensor, and they are ranked:** (1) **forward camera — mandatory**; (2) radar, forward + rear-corner; (3) ultrasonic; (4) IMU / GNSS; (5) **2D lidar — optional**. A behaviour may *require* only the camera |
| D9 | **The island carries its own communication module to the remote command and control centre (RCC)** — a cellular modem on the island's harness and power rail — **and a local control port** for a joystick, wired or short-range radio. Human-driven B5 and B6 run on them; stage-1 design in §6.6 |
| D10 | **Two design branches** share the design and differ in the island's hardware: **Least-Cost** — camera, 5 ultrasonic, 2D lidar, IMU / GNSS on an S32Z2, TC4x or Stellar, no radar; **Maximum-Safety** — camera, 4 radars, 5 ultrasonic, IMU / GNSS on an R-Car X5H (§9.5) |

**Takeaway.** Part 1 answers one question: **which ROS nodes execute where, and which new nodes each side needs.**

# Two engineering facts, not assumptions

### Already ported

`autoware-safety-island` (branch `nano-ros`) runs the vendored Autoware **MPC lateral + PID longitudinal** controllers as standalone firmware on Zephyr / FreeRTOS.

- Subscribes: trajectory, odometry, steering report, acceleration, operation-mode state
- Publishes: `control_cmd`
- Wire-compatible with *unmodified* ROS 2 Humble Autoware
- Closed loop at $\geq$ 19 Hz against a 10 Hz trajectory
- 250-point trajectory bound, `KEEP_LAST` depth 1 on every input

### Already measured

Six-node RT-evaluation island from the Autoware vehicle-interface subsystem, FreeRTOS / zenoh-pico.

- Control cadence held *through a full Linux outage*
- Mean period **9.96 ms**
- 500 ms watchdog tripped in **0.49–0.67 s**
- MRM ramped the vehicle to standstill

**Takeaway.** The allocation is built so the existing island grows into the target island **by addition, not by rewrite**.

What the island must do

Four behaviours, declared before anything is allocated — everything after this exists to deliver them.



# Six declared behaviours

This is the **normative** part of Part 1. Every allocation decision that follows must be traceable to one of these six.

| ID | Behaviour | Statement |
|---|---|---|
| B1 | **Degrade the ADS** when the ODD cannot be held | **Require the ADS to move to an operating domain it *can* satisfy** — progressively constraining it, and demanding it constrain itself, until it does — **before any authority is taken** |
| B2 | **Keep the vehicle** in its lane after HPC loss | When the HPC is gone and B1 therefore *cannot* be used, **take authority and hold the lane** for a bounded 8–15 s window, first down to the **safe speed of the current ODD** (∼60 km/h highway, ∼30 km/h city), then at a **decaying** speed, while staging the exit — pending the HPC's return, else B3 or B4 |
| B3 | **Pull over** if the ODD continues to fail | When degradation does not restore the envelope within a declared deadline, bring the vehicle **out of the running lane** to a standstill in a refuge — nominated by the HPC where one is offered, **found by the island's camera** — lane lines and pull-over space — where one is not |
| B4 | **Emergency stop** when the island is blind | When the sensing that B2 and B3 *both* depend on — the **camera** — is gone, **decelerate to a standstill where the vehicle already is** — no lane keeping, no refuge, no lateral manoeuvre |
| B5 | **Tele-operation** from the remote command centre | A **remote driver at the RCC** drives over the island's communication module, on sensor data the island **streams** — camera, radar, ultrasonic — with the island's guards still armed; the island stops the vehicle itself if the link fails. *Stage 1: from standstill, $\leq 20$ km/h* |
| B6 | **Manual joystick** without streamed sensing | An operator **within sight of the vehicle** drives it by joystick, **wired or short-range radio**, with **no sensor stream**; creep speed, stop on release or link loss. *Stage 1: from standstill, 5 km/h* |

**Takeaway.** A one-way ladder with two exceptions: B1 needs a live HPC, B2 replaces it when there isn't one, B3 ends the drive — and only B2→B1 and B1→ADS_ACTIVE climb back up. B4 is not a rung of that ladder: it is the **floor underneath it**. Two behaviours stop the vehicle, not one: B3 places the stop, B4 cannot. And B1 is the only one with an **upward** exit. B5 and B6 are **human-driven**: the only exits from SAFE_STOP.

# Each behaviour, defined

| | B1 degrade | B2 keep lane | B3 pull over |
|---|---|---|---|
| **Trigger** | ODD margin falls; sensor degradation; HPC declares its own ODD exit | **HPC unavailable** (N2) while the lane is still observable by the camera — i.e. B1 has no executor | B1's $T_{odd}$ expired with the HPC alive; or B2's $T_{hold}$ expired with the camera still seeing the lane — entered at $v_{entry}$ |
| **Authority** | **Island constrains, HPC executes.** No authority taken | **Island steers and brakes.** Nominal follower deselected, no HPC in the loop | **Island steers, brakes, signals** — the only behaviour where it initiates lateral *displacement* |
| **Response** | notify → **demand an ODD switch** → restrict ($v_{max}$) → request MRM. The cap is held *underneath* the switch, not instead of it | Track the camera's lane lines (map corridor refines while fresh; no heading tube); slow to $v_{odd}$, then the $v_{hold}$ **ramp**, hazards on | Search **both** refuge sources — the camera finds lane lines and space → hazards on at entry → signal → displace → decelerate → secure |
| **Deadline** | $T_{ack}$ 2 s to declare · $T_{odd}$ 10 s to be in it · 3 s comply · 10 s begin · 60 s complete | Transfer ≤ 70 ms; hold at most $T_{hold}$ = 8–15 s — **finite**; at $T_{hold}$ the vehicle is already at $v_{entry}$ | $T_{refuge}$ 5 s to find a refuge; then 10 s begin · 60 s complete |
| **Observable** | `/si/out/odd_verdict`, `/si/out/constraints`. **MrmState stays NORMAL** | N3 mode `LANE_KEEP_HOLD`, corridor source, deviation, hold budget left | `MrmState = MRM_OPERATING`, behaviour `PULL_OVER` |
| **On failure** | Non-compliance is a fault; a domain declared but not entered is a *false declaration* — a SOTIF event at N11 | Camera lost → B4 at once; AEB or actuator fault → emergency stop | Abort ladder → in-lane stop (B2 authority) → emergency stop. Search *inconclusive* at $T_{refuge}$ → B4 |
| **Exits** | **Up:** a domain that passes both acceptance tests → **normal L4 operation, in a narrower domain**. **Down:** $T_{odd}$ expires → blackout → B4, HPC lost → B2, otherwise → B3 | **Up:** HPC re-qualified over a confirmation window → B1 at `CONSTRAIN`. **Down:** $T_{hold}$ expires → B3 if the camera sees the lane, else B4 — nothing else | Standstill in refuge → `SAFE_STOP`. **Terminal** — exactly as B4 is |
| **Min. sensor rung** | **B** — radar-vs-camera divergence *and* camera contrast; friction needs **C** | **A** — camera + chassis signals. Highway $v_{odd}$ needs **B** | **A** at ≤30 km/h; **B** above (rear-corner radar) |

# B1 — the ODD switch: what the island demands, and what it verifies

new in 20260918

Read as a constraint ladder, B1 is a demand that the ADS *slow down*. That is only its **enforcement**. Its object is an **operating domain**.

| # | Who | What happens |
|---|---|---|
| 1 | island N19 | Declares the current domain **unsatisfiable**: `odd_satisfiable = false` + the measured envelope + the dimension that failed. **Names no target domain** |
| 2 | HPC | **Selects** a domain it can meet. This is planning, and it happens *entirely* on the HPC |
| 3 | HPC H4 | Declares the new `active_odd` and its bounds — within $T_{ack}$ = 2 s |
| 4 | island N19 | **Verifies the declaration**: those bounds must lie inside the island's own measured envelope |
| 5 | island N2 | **Verifies the behaviour**: measured speed, acceleration and curvature converge on them — within $T_{odd}$ = 10 s |
| 6 | island N3 | Accepts, and releases the cap to ALLOW |

**The cap does not come off for the declaration. It comes off for the behaviour.** Between 1 and 6 the vehicle is held at $v_{max}$ whatever the HPC has declared, so a false declaration buys the HPC no speed — only time, and only $T_{odd}$ of it.

## The asymmetry is the point

The island holds no catalogue of the HPC's operating domains, no model of which L4 behaviours each admits, and no means of choosing between them — all three are planning, which R4 and R5 keep off the island.

But a *measurement* is enough to **reject** an answer without being able to **produce** one.

## Steps 4 and 5 are separate on purpose

Step 4 catches an HPC reporting a domain it *could not* be in. Step 5 catches one reporting a domain it *is not* in — the more dangerous failure, because the declaration is plausible. Passing 4 and failing 5 is a **SOTIF event**: §6.4's confidently-wrong case, on a second channel.

# B1 — the only behaviour with an *upward* exit

new in 20260918

B3 and B4 end the drive cycle; B2 holds one open. B1 is the only behaviour that returns the vehicle to **normal L4 operation** — in a narrower domain than the one it started in, driven by the HPC, with the island back to supervising. That narrowing is B1's *outcome*, not a residual fault.

| At $T_{odd}$ expiry | Takes over | Why |
|---|---|---|
| Sensing blackout (N17) — camera lost | B4 | Capability, not situation. Neither B2 nor B3 has a precondition left |
| HPC unavailable (N2) | B2 | The executor of every B1 rung is gone, and the corridor is still observable |
| HPC alive, non-compliantor out of satisfiable domains | B3 | The situation has not been recovered and will not be. The drive ends in a refuge |

**Honest exhaustion is not non-compliance — it does not reach this table at all**

"None of my domains is satisfiable" is a **compliant** answer, not a timeout. It routes straight to **Request MRM** — an HPC-executed pull-over, with the map, the traffic law and the full planner available to it, which is strictly better than the island-executed one under B3. $T_{odd}$ bounds **silence and non-compliance**, not honesty.

**Takeaway.** The successor is chosen by what the island still **has** — never by what the HPC did wrong, because the island cannot always tell a non-compliant HPC from a dead one, and does not need to. The tests run **in that order**: capability loss, then executor loss, then situation loss — the order in which each removes the preconditions of the options below it.

# B2 — first down to the ODD's safe speed, then a fork

The island's first act on taking authority is to stop driving at a speed only the HPC could justify. $v_{odd}$ is the safe speed of the ODD the vehicle is in — its road class latched from H4's last `active_odd`; stale or missing selects the most restrictive class.

| Class | Entry | $v_{odd}$ | To $v_{odd}$ at 2 m/s² | Left of 8 s / 15 s |
|---|---|---|---|---|
| Highway | 100 km/h | 60 km/h | 5.6 s, 124 m | 2.4 s / 9.4 s |
| City road | 50 km/h | 30 km/h | 2.8 s, 31 m | 5.2 s / 12.2 s |

A 2–5 s window would spend itself on the descent alone. **8–15 s** contains it and still leaves the HPC a real chance to return. Then $v_{hold}$ ramps on to $v_{entry}$.

## At $T_{hold}$: exactly two successors

Tested capability-first:

**camera lost, or lane lines lost** → B4

**camera still sees the lane** → B3, entered at $v_{entry}$ with the refuge search already running

No third exit. The in-lane stop is reached only through B3's search. Losing the camera mid-hold applies the first test early.

## The lane is the camera's

The camera is the required corridor; the map corridor only refines it while fresh. The heading tube is withdrawn: a lane the island cannot see is handed to B4. Minimum rung **A**; the highway $v_{odd}$ needs radar, rung **B**.

# B4 — the floor: emergency stop when blind

new in 20260903

| | |
|---|---|
| **Trigger** | N17 declares **sensing blackout** — the **camera** invalid or the lane lines lost, whatever radar still reports. Equivalently: B2 has no corridor *and* B3 no verifiable refuge. Also N9 actuator fault, or I19 AEB |
| **Precondition** | **None, deliberately.** A fallback whose entry can be blocked is not a fallback |
| **Authority** | **Brakes, and declines to steer.** No corridor, so nothing to steer against; steering held under N7's rate limit, relaxing to straight |
| **Response** | **Hazards on at entry** → I6 commands directly → $a_{emg}$ ramp → I5 secures. **No trajectory is planned** |
| **Deadline** | Transfer $\leq$ 70 ms. $a_{emg}$ is **open decision 15** |
| **Observable** | `MrmState = MRM_OPERATING`, behaviour `EMERGENCY_STOP` — the one value the existing enum carries |
| **Exits** | `SAFE_STOP` **only** — no edge back up. Terminal even if sensing returns mid-stop |
| **Min. sensor rung** | **0 — none.** Chassis signals only |

## Why stopping in the lane is accepted

The argument is *comparative*. The only other things a blind island can do are continue on stale evidence, or attempt a manoeuvre it cannot verify — both worse, and both failing **unboundedly**. A stop is bounded by the vehicle's own kinetic energy.

The residual risk is real: it belongs in the HARA as a **named accepted risk**.

## B4 is not "AEB fired"

I19 is a *reflex* against one detected obstacle, presupposing a working forward sensor. B4 is a *state*: the island no longer knows what is ahead at all. AEB escalates *into* B4; the reverse is impossible.

# B5, B6 — a person drives

new in 20261023

| | B5 tele-operation | B6 manual joystick |
|---|---|---|
| **Who drives** | Remote driver at the **RCC** | Operator **within sight** of the vehicle |
| **Link** | Island's cellular module, N21 | Wired or short-range radio, N23 |
| **Sensing** | **Streamed** by N22: camera (required) · radar (rung B) · ultrasonic (rung C) | **None streamed** — the operator's own eyes |
| **Entry** | Standstill: RCC request granted by N3 — camera valid, link in bounds, vehicle confirmed | Standstill, or B5 stopped first — **local pre-empts remote**; enabling switch held |
| **Speed** | $v_{teleop} = \min(20\,km/h, v_{vis})$, from the measured latency | 5 km/h forward, 3 km/h reverse |
| **Guards** | I19 · I20 · N10 · N9 stay armed | I19, I20 where sensing is valid |
| **Link lost** | $T_{link}$ = 300 ms, or a command > 1.0 s old → in-lane stop if the camera sees the lane, else B4 | Release, E-stop or $T_{local}$ = 200 ms → immediate stop (I6, I5), ≈ 0.85 m |
| **Exit** | Released at standstill → SAFE_STOP; the RCC may then re-engage the ADS | Released at standstill → SAFE_STOP only |
| **Observable** | `OperationModeState = REMOTE` | `OperationModeState = LOCAL` |
| **Min. rung** | **A** — the camera the driver watches | **0** — none |

## The communication module (D9)

Three new nodes, outside the counts until the WG accepts the design:

N21 `si_rcc_link` — RCC session, authentication, heartbeat, latency, E2E on every remote command

N22 `si_teleop_streamer` — camera, radar, ultrasonic to the RCC; own low partition, below P2

N23 `si_local_control` — joystick port, pairing, dead-man, creep limit

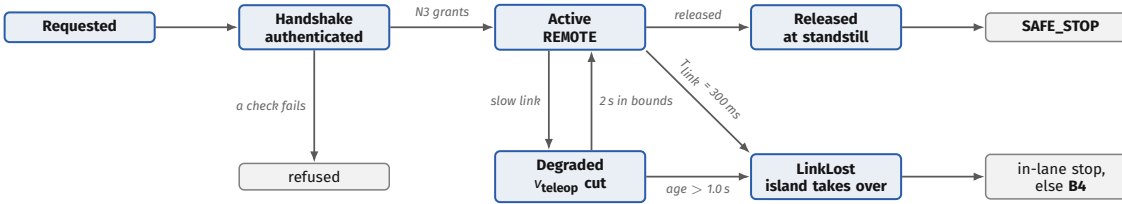
The link ends **on the island**, so B5 survives the HPC.

## Still open — decisions 19, 20

Ratifying the stage-1 numbers · re-engaging the ADS · modem redundancy and key ownership · stage 2: moving take-over, path-based assistance · joystick certification and radio band.

# B5 — the session, and the frame-stamp echo

new in 20261023




- **The grant**: standstill · camera valid (N17) · round-trip time and stream age in bounds · no B6 · the driver confirms the vehicle's identity, overlaid by the island.
- **The frame-stamp echo**: every command carries the island's capture time of the frame the driver was looking at. The island measures, **on its own clock**, how old the picture behind each command was — capture, uplink, screen, reaction and downlink in one number. No shared time base is needed.
- **The command**: steering and signed speed *setpoints* at 50 Hz, with a dead-man flag — into `I1`'s remote source, validated by `I8` and `N10` like any other.

# B5 — how fast, and when the island stops

new in 20261023

The vehicle must stop inside the free road **the island itself confirms**, after the whole loop:

$$v_{\text{teleop}} = \min\left(20\ \text{km/h},\ v_{\text{vis}}\right), \qquad v_{\text{vis}}\, t_{\text{loop}} + \frac{v_{\text{vis}}^2}{2\, a_{\text{tele}}} = d_{\text{free}} - 2\ \text{m}$$

$t_{\text{loop}}$ = measured command age + island + actuator; design value 1.8 s (0.23 capture-to-screen, 1.2 reaction, 0.12 downlink, 0.04 island, 0.2 actuator) · $a_{\text{tele}}$ = 3 m/s² · $d_{\text{free}}$ from N16, capped by what the stream can show

| $d_{\text{free}}$ | 10 m | 15 m | 20 m | 30 m | 40 m |
|---|---|---|---|---|---|
| $v_{\text{vis}}$ at 1.8 s | 12 km/h | 18 km/h | 23 km/h | 31 km/h | 38 km/h |

| Check on every command | Bound | On failure |
|---|---|---|
| Session, sequence, MAC, CRC (SI-E2E/R) | exact; replay window 64 | drop; security event to N8 |
| Dead-man | every command | speed to zero, brake held |
| Command validity | 100 ms | setpoint decays to zero |
| Command age (frame-stamp echo) | 600 ms soft · 1.0 s hard | Degraded, $v_{\text{teleop}}$ cut · LinkLost |
| Heartbeat | $T_{\text{link}}$ = 300 ms | LinkLost → in-lane stop, else B4 — ≈ 7 m from 20 km/h |

Takeaway. **A slow link makes the vehicle slower before a lost link makes it stop.** Least-Cost confirms 15–20 m: about 18–23 km/h, forward only. Maximum-Safety runs at the 20 km/h cap and may reverse at creep.

# B6 at the vehicle, and the RCC link's third profile

new in 20261023

## B6 — a cableless-machinery joystick

- Three-position **enabling switch** and a latching **emergency stop**; IEC 62745, stop functions at PL d
- By cable, or a radio receiver on the **same CAN FD port** into N23
- Pairing needs a button on the vehicle, at standstill; one device at a time
- **5 km/h** forward, **3 km/h** reverse, 0.5 m/s²
- Release, E-stop or **200 ms** of silence → stop through I6, I5: ≈ 0.85 m from 5 km/h
- E-stop also **hardwired** to the actuator gateway
- Never hands back to the ADS

## SI-E2E/R — the cellular link is an *open* network

| | |
|---|---|
| Safety | Data ID 16-bit · counter **32-bit**, replay window 64 · length · CRC-32 |
| Age | the frame-stamp echo — island clock only |
| Session | 32-bit ID, one operator, one vehicle |
| Security | MAC from mutually authenticated **TLS / DTLS 1.3**; keys in the processor's HSM |
| Video | own SRTP channel; its age is what matters |

A CRC does not stop an attacker who can compute one (EN 50159): a security failure and a safety failure are **both** raised to N8, and repeated failures of either count as a link loss.

# Four episodes, a floor — and two human exits

## Precedence

- B1 is the **first response**, and it is cooperative: the HPC stays in control. It therefore *requires a working HPC*.
- B2 **replaces B1 when that requirement fails**. Same intent — buy time rather than commit — but the authority is inverted: the island steers. Admissible only while the camera sees the lane; bounded by $T_{\text{hold}}$ (8-15 s), first down to the ODD's safe speed, then **forks**: B3 if the camera still sees the lane, B4 if not.
- B3 is **terminal for a worsening *situation***: B1 exhausted its deadlines, or B2's hold expired with the HPC still gone.
- **Two behaviours end the drive, not one.** B3 is a **placed** stop — refuge verified; B4 is an **unplaced** one, promising a standstill and nothing about *where*. Both terminal, both to SAFE_STOP, two HARA entries.
- **One-way except two edges back up:** B2 → B1 — a re-qualified HPC gets authority back at CONSTRAIN; and B1 → ADS_ACTIVE — the switched ODD passed both acceptance tests and held, so the HPC drives normally in the narrower domain.
- B4 is **underneath all of them**, not after them. B1–B3 escalate on a worsening *situation*; B4 on a loss of *capability*, which can go at any rung. **Every state has an edge into it; its only edge out is SAFE_STOP.**
- B5 and B6 sit **outside the ladder**: they are how a **person** takes the vehicle back — the only edges out of SAFE_STOP. Both return there when released; on a lost link B5 falls to the in-lane stop or B4, B6 straight to a stop. B6 pre-empts B5.

### B3 deliberately leaves the lane

A pull-over leaves the running lane *on purpose*. The containment check N5 must be **suspended under authorisation** for the duration and re-armed against the refuge boundary instead.

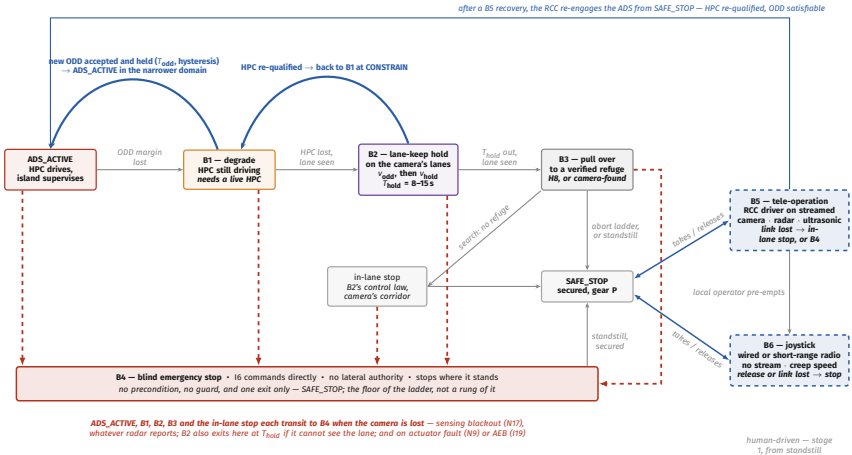
An unsuppressed containment check would abort every pull-over it was asked to protect.

Authorisation comes from N3 and nothing else, bounded by N7, recorded by N11.

### Containment supervision is *not* B2

The continuous N5 check *while the HPC steers* is a *validate* function — its output is a veto and a fault, not a behaviour. B2 is what the island does when it must produce the steering itself.

# The ladder, drawn




**Takeaway.** Two edges climb: B2 → B1 when the HPC is re-qualified, and B1 → ADS_ACTIVE when the switched ODD is verified and held. B4 is the **floor**, not a fourth rung: every state above has an edge into it, and its only edge out is to SAFE_STOP — never back up the ladder. Out of SAFE_STOP, only a person: B5 or B6.

# Behaviour-to-node traceability

**A node that serves no declared behaviour does not belong on the island**

| | Detect | Decide | Act |
|---|---|---|---|
| B1 | N19 ODD · N17 health · N16 world model | N10 envelope · N3 · N8 | N1/H1 constraint egress → HPC re-plans |
| B2 | N2 HPC supervisor · N14 camera lanes (required) · N17 · N6 · H3 corridor | N3 hold budget + hand-back + B3/B4 fork · N5 corridor · I8 · N10 | I13 pure_pursuit + N7 → I1 → I16; I3 for hand-back |
| B3 | N16 free space · N6 rear radar · H8 refuge · N14 camera: lane lines, road edge, space | N18 · I7 · N3 | N7 → I13 → I1 → I16; I5 secures |
| B4 | N17 blackout (camera lost) · N16 no corridor · N9 · I19 | N3 (EMERGENCY_STOP, no exit) · N8 · I7 | I6 → I1 → I16; I5 secures. **No exteroceptive input** |
| B5 | N21 RCC link · N17 camera validity · N16 | N3 grants, revokes · mode REMOTE | N21 → I1 remote source → I16; N22 streams to the RCC. *Pending* |
| B6 | N23 local port — pairing, dead-man, heartbeat | N3 · mode LOCAL, pre-empts REMOTE | N23 → I1 local source → I16. *Pending* |

**Takeaway.** HPC loss does not commit the vehicle to a terminal manoeuvre — it commits it to a *bounded* one. When it does become terminal, it is B3 or B4, and both raise the hazard lamps at entry.

How the allocation was decided

Eight rules, one of them decisive: what must still work after the HPC is gone.



# Eight partitioning rules

| Rule | Statement | Consequence |
|---|---|---|
| **R1** Survivability | If the function must still work *after the HPC is gone*, it runs on the island | Forces the whole final-command path onto the island |
| **R2** No orphan inputs | An island node must not depend on an HPC-only topic without a defined safe behaviour when it goes silent | Forces island-local state estimation and fault sources |
| **R3** Monitor independence | A monitor must not share a failure mode with what it monitors | HPC liveness supervision is island-resident |
| **R4** Bounded resources | Static memory, bounded WCET, no dynamic discovery | Excludes map-dependent and learning-based nodes from the ASIL tiers |
| **R5** Narrow interface | Typed, fixed-rate, rate-limited, E2E-protected | Keeps HD map, point clouds, images, object lists off the link |
| **R6** Single actuation authority | Exactly one component owns the final command, and it is on the island | The gate is the arbitration point and the ASIL-D anchor |
| **R7** Diversity over duplication | Prefer a *different* sensing channel over a copy of the HPC's | A separate low-resolution sensor set (D7) |
| **R8** Certifiable critical set | The ASIL-D set must be small enough to argue about | Two-tier controller |

**Takeaway.** **R1 is the decisive one.** Everything else falls out of asking what must survive the HPC.

# Criticality tiers — and the staleness rule

| Tier | Integrity | Contents | Cadence |
|---|---|---|---|
| T0 | ASIL-D | Command gate, state machine, watchdog, emergency-stop MRM, actuator TX, E2E gateway, island odometry | 100 Hz |
| T1 | ASIL-B(D) | Control validator, fallback controller, safety envelope, corridor monitor, actuator supervisor | 50 Hz |
| T1s | ASIL-B(D) | Island sensor ingest, safety world model, emergency braking, near-field collision detection | 20–50 Hz |
| T2 | QM, monitored | MPC lateral + PID longitudinal follower, raw vehicle command converter, shift decider | 33–50 Hz |
| T3 | QM | Event recorder, UDS / DTC diagnostics, telemetry egress | 1–10 Hz |

### ASIL decomposition by monitoring

The MPC and its OSQP solver **never need to be ASIL-D**: T0 bounds every command it produces and can replace it within one control period.

### The staleness rule for sensing

**T0 never blocks on T1s.** Stale or invalid world state → most conservative assumption, blind deceleration MRM. *Perception improves the fallback; it is never a precondition for it.*

# Reminder — what QM and ASIL-A … D actually mean

| Level | What it demands | HW architectural metrics | Where it appears in this design |
|---|---|---|---|
| **QM** | **No** ISO 26262 safety requirement — developed under a normal quality system | — | `T2` MPC + PID follower (*monitored*) · `T3` recorder, UDS, telemetry |
| **ASIL A** | Lowest integrity. Full 26262 lifecycle, lightest verification and confirmation measures | not specified | Not used — nothing on the island lands here |
| **ASIL B** | Systematic rigour on requirements, design and test; review independence begins to bite | SPFM $\geq$ 90 % · LFM $\geq$ 60 % · PMHF $<$ 100 FIT | `T1` validator, envelope, corridor, actuator supervisor · `T1s` sensing, AEB |
| **ASIL C** | As B, with tighter metrics and more independent verification | SPFM $\geq$ 97 % · LFM $\geq$ 80 % · PMHF $<$ 100 FIT | Not used — the decomposition below skips D to B |
| **ASIL D** | Highest. Strictest process, independent confirmation, demonstrated *freedom from interference* | SPFM $\geq$ 99 % · LFM $\geq$ 90 % · PMHF $<$ 10 FIT | `T0` command gate, state machine, MRM, E2E gateway, island odometry |

SPFM = single-point fault metric · LFM = latent fault metric · PMHF = probabilistic metric for random hardware failures · 1 FIT = 1 failure per $10^9$ h

### Where the letter comes from

The HARA assigns an ASIL to each *hazardous event* from **S**everity (S0–S3) × **E**xposure (E0–E4) × **C**ontrollability (C0–C3); **D** is the corner case S3 + E4 + C3 — life-threatening, likely encountered, not driver-controllable. The ASIL attaches to the **safety goal**, never to a box: components inherit it *by allocation*, which is exactly what this deck is doing.

### Reading ASIL-B(D)

*ASIL decomposition*, ISO 26262-9: one ASIL-D requirement may be split across two **sufficiently independent** elements, each built to **B** rigour while still carrying a **D**-derived requirement — written B(D). The parenthesis is a debt, not a discount: lose freedom from interference and it collapses back to D.

**Takeaway.** A decomposition is only as strong as the independence argument underneath it.

# The same levels, in examples

| Level | A familiar vehicle function | Why it lands there | Where it appears in this design |
|---|---|---|---|
| **QM** | Infotainment screen freezes · seat-memory recall · cabin ambient lighting | **S0** — no injury path at all | T3 event recorder, UDS export, telemetry. *The* T2 *follower is QM only because* T0 *bounds it* |
| **ASIL A** | Rear position lamps fail after dark | S1 · E4 · C3 | Nothing — no island function has a *light-injury* worst case |
| **ASIL B** | Brake lamps or low beam fail | S2 · E4 · C2 | T1 I8 validator, N10 envelope, N5 corridor · T1s N16 world model, I19 AEB — all B(D) |
| **ASIL C** | Partial, unintended loss of steering assist at speed — still steerable, with effort | S3 · E4 · C2 | Nothing — the decomposition goes straight from D to B(D)+B(D) |
| **ASIL D** | Unintended airbag deployment · loss of braking · unintended steering to lock | S3 · E4 · C3 | T0 I1 gate, N3 state machine, N2 supervisor, I6 emergency stop, N1 E2E, N4 odometry |

S/E/C values are *illustrative* — the real ones come from the vehicle manufacturer's HARA, not from the standard. Anchor point: S3 + E4 + C3 = D, and every step down of one parameter drops one level.

**Why an L4 island is nearly all D**

In L2, part of the controllability argument is carried by the driver, so **C2** is defensible and an S3 hazard lands at **C**. Take the driver out and the same hazard is **C3** — nobody is left to catch it — and it lands at **D**.

The island's ASIL profile is not conservatism. It is the arithmetic of having no fallback-ready user.

**One box, two letters**

An ASIL belongs to a *hazard*, not to a component, so one node can sit at two levels at once. N16 is B(D) for making the MRM obstacle-aware — but "*a stale N16 must never stall the MRM*" is a **D** requirement, and it is discharged in T0 by the staleness rule, not in N16.

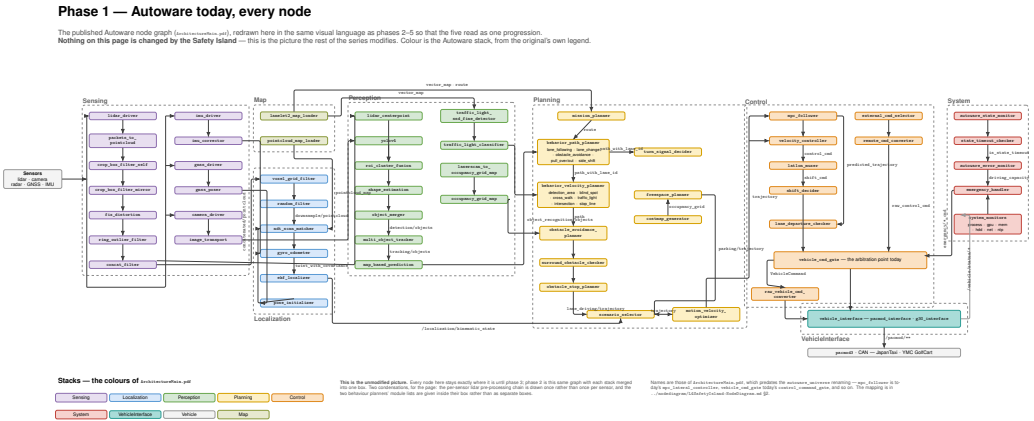
**Takeaway.** Every letter is a property of a *hazard*, never of a box — and with no driver left to catch it, the same hazard earns a bigger letter.

The allocation

Eighteen Autoware nodes move.
The interesting part is the
three that were not obvious.

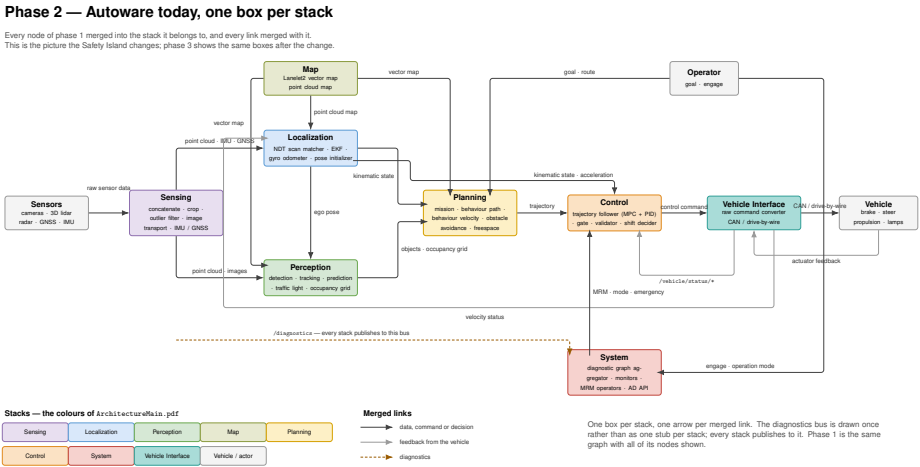


# Phase 1 of 5 — Autoware today, every node



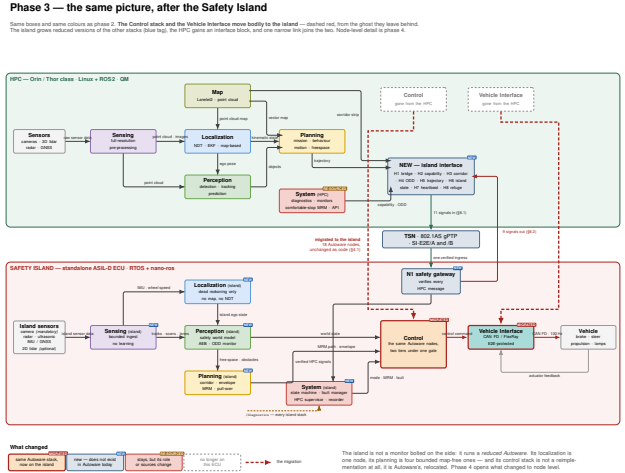

**The starting point** — nothing on this slide is changed by the design. It is the picture the next four phases modify.

# Phase 2 of 5 — the same graph, one box per stack



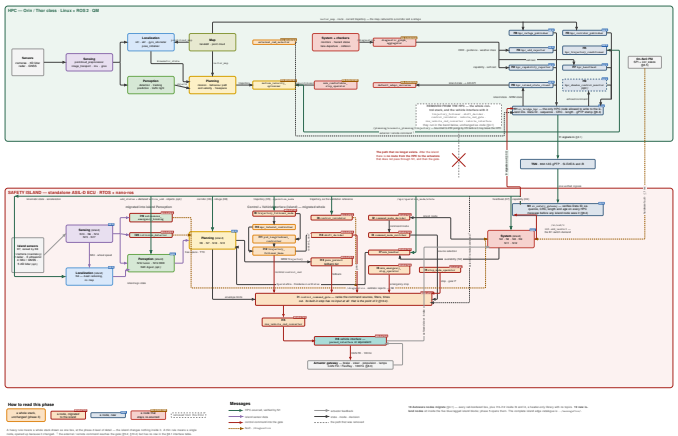

**The detail folded away** — every node of phase 1 merged into its stack, and every link merged with it.

# Phase 3 of 5 — the same boxes, after the island



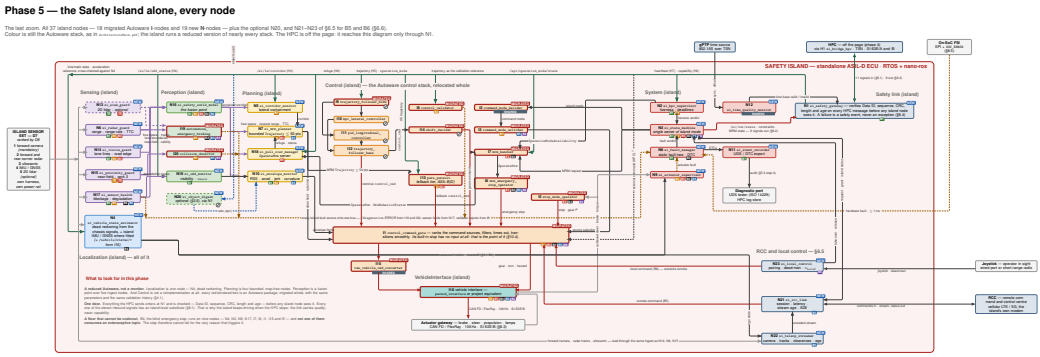

**The change itself** — Control and the Vehicle Interface migrate bodily; the island grows reduced versions of the other stacks; one narrow link joins the two.

# Phase 4 of 5 — what changed, at node level



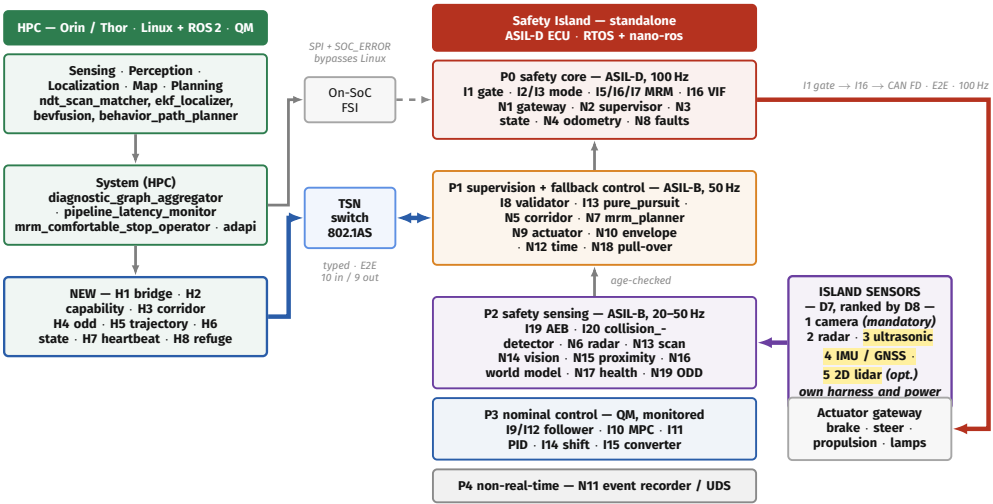

**The delta, opened up** — on the HPC only what was removed, re-sourced or newly designed; on the island only what migrated. Everything else stays at the phase-3 level.

# Phase 5 of 5 — the Safety Island alone




**What the island then contains** — all 37 nodes: the 18 migrated Autoware nodes and the 19 new ones. The HPC reaches this picture only through **N1**.

# Node allocation across the two domains




**Two sensor sets, two harnesses, two power rails** — so the only paths that survive HPC loss originate inside the island boundary.

# Autoware nodes on the island — authority, mode, MRM

| # | Status | Package | Function on the island |
|---|---|---|---|
| I1 | ❍ | autoware_control_command_gate | **The arbiter.** Selects among command sources, filters abnormal values, generates the built-in stop on source timeout |
| I2 | | autoware_command_mode_decider | Decides which source holds authority. Plugin logic replaced by a static, table-driven decider |
| I3 | | autoware_command_mode_switcher | Applies the decided mode; drives transitions |
| I4 | | autoware_command_mode_types | Shared type and constant library; header-only, shared with the HPC |
| I5 | | autoware_stop_mode_operator | Stop commands, optional auto-park. **Does not depend on localization** — reads vehicle status only |
| I6 | ❍ | autoware_mrm_emergency_stop_operator | Emergency-stop MRM; publishes control commands directly |
| I7 | | autoware_mrm_handler | Selects the MRM behaviour. Must be island-resident: its reason to act *is* HPC loss |
| I16 | ❍ | *vehicle interface* | CAN FD / FlexRay to the actuator gateway. Vehicle-specific, E2E-protected |
| I17 | | autoware_operation_mode_transition_manager | *Legacy path only* — superseded by I2/I3 |
| I18 | ❍ | autoware_vehicle_cmd_gate | *Legacy gate* — alternative to I1. Do not deploy both |

✓ vendored and running in `autowarefoundation/autoware-safety-island` · ❍ demonstrated on the RT-eval island only, ASI adoption pending · blank = not started.

*Checked against the upstream repository, 2026-08.*

The upstream ASI repository vendors **no** gate, mode, MRM or vehicle-interface package — its whole surface is the trajectory follower (next slide). The ❍ marks come from the separate RT-eval island, which runs *a* command gate, an MRM operator and the 100 Hz actuator path (FreeRTOS · zenoh-pico). Whether that gate becomes I1 or I18 on the target island is **open decision 1** — hence the mark on both.

**Takeaway.** **Not one node on this slide exists in the ASI repository today** — this is phase 1 in its entirety. The target set separates *decide* / *switch* / *gate* into three nodes, exactly the decomposition a safety argument needs.

# Autoware nodes on the island — control, sensing, vehicle

| # | Status | Package | Function on the island |
|---|---|---|---|
| I8 | | autoware_control_validator | Checks the control command against trajectory and vehicle limits |
| I9 | ✓ | autoware_trajectory_follower_node | Host node for the two controller plugins |
| I10 | ✓ | autoware_mpc_lateral_controller | Nominal lateral control. OSQP iteration cap + time budget still required |
| I11 | ✓ | autoware_pid_longitudinal_controller | Nominal longitudinal control |
| I12 | ✓ | autoware_trajectory_follower_base | Controller plugin base |
| I13 | | autoware_pure_pursuit | **Fallback lateral control** — closed-form, bounded time. New to the island |
| I14 | | autoware_shift_decider | Derives the gear command |
| I15 | | autoware_raw_vehicle_cmd_converter | Generic accel/steer to vehicle-specific pedal/torque. Must sit *downstream of the gate* |
| I19 | | autoware_autonomous_emergency_braking | **Island AEB** — moved from the HPC by D7. Point cloud + IMU ego path |
| I20 | | autoware_collision_detector | **Near-field guard** — moved from the HPC by D7. Point-cloud-only, ∼5 m footprint |

✓ vendored and running in `autowarefoundation/autoware-safety-island` · ❍ demonstrated on the RT-eval island only, ASI adoption pending · blank = not started.

*Checked against the upstream repository, 2026-08.*

Those four nodes do not travel alone. The repository also vendors `autoware_msgs`, `autoware_osqp_interface`, `autoware_universe_utils`, `autoware_motion_utils`, `autoware_interpolation` and `autoware_vehicle_info_utils` — six support libraries, not nodes, but the real width of the port. Pinned to **Autoware 2025.02 / Universe 0.40.0**, CycloneDDS 0.11.x, Zephyr 3.6.0; four runtime targets (`zephyr-fvp`, `zephyr-s32z`, `freertos-posix`, `freertos-s32z2`). Interface: **5 subscriptions** — trajectory, odometry, steering report, acceleration, operation mode — **and one publication**, `control_cmd`.

**Takeaway.** **18 Autoware nodes on the target path; 4 of them are running today** — the vendored MPC + PID follower, unmodified, closed loop against stock Autoware. The other 14 are additions to that image, not rewrites of it.

# What stays on the HPC — and why

**Whole stacks, HPC only:** `sensing`, `perception`, `localization`, `map`, `planning`. They fail R4 (unbounded memory, learning-based, map-sized state) and R5 (their traffic would flood the link).

| Package | Reason |
|---|---|
| `autoware_spheric_collision_detector` | Object-list based. Its point-cloud siblings moved to the island |
| `autoware_obstacle_collision_checker`, `autoware_predicted_path_checker` | Depend on predicted paths and map |
| `autoware_lane_departure_checker` | Requires the Lanelet2 vector map. Island equivalent is N5, dual-sourced from H3 and N14 |
| `autoware_external_cmd_selector`, `autoware_joy_controller` | Remote and manual sources. They reach the island as *one* E2E-protected command source into the gate. Under D9 the B5/B6 paths end on the island: candidates to migrate |
| `autoware_mrm_comfortable_stop_operator` | Publishes a `VelocityLimit` to the **planner** — it cannot function without planning |
| `autoware_diagnostic_graph_aggregator`, `autoware_hazard_status_converter` | Large config-driven fault trees. Island runs its own static fault manager N8 |
| `autoware_system_monitor` and the eight other monitors | They monitor HPC resources. Under R3 they are *evidence sources*, not the safety monitor |
| `autoware_default_adapi_universe` | API and orchestration surface |

**Takeaway.** Consequence to handle: the island's controllers subscribe to `/localization/kinematic_state` and `/localization/acceleration`, **both HPC-produced**. Resolved by N4.

# Design move 1 — a two-tier controller

D2 puts “the control module” on the island. Taken literally that places an iterative QP solver inside the safety-critical ECU, and the certification argument then has to cover OSQP’s worst-case execution time. **That argument is expensive and fragile.**

| | Nominal tier T2 | Fallback tier T1 |
|---|---|---|
| Lateral | `autoware_mpc_lateral_controller` | `autoware_pure_pursuit` |
| Longitudinal | `autoware_pid_longitudinal_controller` | Fixed deceleration profile N7 |
| Integrity | QM, monitored | ASIL-B(D) |
| Time budget | Soft — may be dropped | Hard — always meets deadline |
| Switch trigger | MPC budget overrun · validator reject · solver non-convergence · tracking error above bound | |

**The behaviour is already live**

The existing port commands an emergency stop on excessive tracking error. This formalises that escalation and inserts a *graceful* intermediate step instead of going straight to a stop.

# Design move 2 — two Autoware nodes can migrate

Both were assumed to need perception object lists. Their actual interfaces say otherwise, and D7 removes the remaining obstacle.

| | `autoware_autonomous_emergency_braking` | `autoware_collision_detector` |
|---|---|---|
| Perception objects | **Optional** — `use_predicted_object_data=false` | **Optional** — `use_dynamic_object=false` |
| Point cloud | `use_pointcloud_data=true` | `use_pointcloud=true` |
| Ego path source | IMU kinematic prediction, *or* the MPC predicted trajectory | none; current footprint only |
| Map dependency | none | none |
| Output | ERROR on `/diagnostics` | ERROR on `/diagnostics` |
| Island sensor | radar tracks → `PointCloud2` (N6); 2D lidar where fitted | 2D lidar + ultrasonic — ultrasonic rank 3, lidar optional: I20 runs from rung C up |

AEB's IMU path comes from N4, its point cloud from N6's radar tracks (N13 where a lidar is fitted), and — usefully — its optional MPC path from I10, *which is already on the island*.

**Takeaway.** The island's emergency braking is not a bespoke reimplementation but **the same Autoware node, same tuning parameters, same validation history** — running on a diverse sensor channel.

# Design move 3 — MRM splits into two classes

| Graded reaction | Implemented by | HPC alive? |
|---|---|---|
| 1 · Notify and request self-recovery | Island → HPC constraint / veto; HPC re-plans | **Yes** |
| 2 · Restrict speed, enlarge margins | `mrm_comfortable_stop_operator` (HPC) + island envelope | **Yes** |
| 3a$_0$ · **Lane-keep hold** (B2) — down to $v_{odd}$, 8–15 s, then 3a′ or 3c | Island: I13 on the camera's lane lines | No |
| 3a · **Obstacle-aware** in-lane stop | Island: fallback tier + N7 along the camera's corridor | No |
| 3a′ · **Pull-over** | Island: N18 to an HPC-nominated refuge or camera-found space | No |
| 3b · Emergency braking on imminent collision | Island: I19 on island sensors | No |
| 3c · Emergency stop (B4) | Island: I6 — blind, no steering, no world model needed | No |
| 3d · Last resort | Island: gate built-in stop I1 | No |
| 4 · Audit | Island N11 + HPC log store | No |

**Takeaway.** Steps 1–2 are cooperative degradation; 3a–3d are island-autonomous. **The island must never depend on step 2 succeeding before it can reach step 3** — and **every rung above 3c has a direct edge into it**, because a guarded floor is not a floor.

Island sensing (D7)

Low resolution, high robustness — and a second view of the road that fails differently.



THE AUTOWARE FOUNDATION

# The island sensor set

Selection principle: **robustness and boundedness over resolution.** Every sensor must degrade gracefully, emit a small fixed-size payload, and be processable without a neural network.

| Sensor (D8 rank) | Link | Data class | Rate | What it gives the MRM |
|---|---|---|---|---|
| **1** Forward camera — *mandatory* | GMSL | Lane lines, road edge, lateral space | 10–20 Hz | **B2**'s lane; **B3**'s lane lines and pull-over space; its loss *is* **B4** |
| **2** Radar, fwd + rear corner | CAN FD | Object tracks: range, range-rate, azimuth | 20 Hz | Range and closing speed in rain, fog, dark; AEB's point cloud; highway $v_{odd}$, B3 above 30 km/h |
| **3** Ultrasonic | LIN / CAN FD | Near-field, 8–12 channels | 10–20 Hz | Standstill, creep, the final metres of a pull-over; tight-to-kerb end from rung C |
| **4** IMU / GNSS | direct / CAN FD | Vehicle dynamics; position | 100 Hz | N4 independent of the chassis bus; friction; GNSS corroborates ODD class |
| **5** 2D lidar — *optional* | 100BASE-T1 | Planar scan, ∼1080 points | 10–40 Hz | Confirms or vetoes only — never qualifies a refuge or supplies a corridor |
| — chassis | vehicle bus | Wheel speed, steering, ESC yaw / accel | 100 Hz | Not ranked: on every vehicle; what rung 0 stands on |

**Two wiring rules**

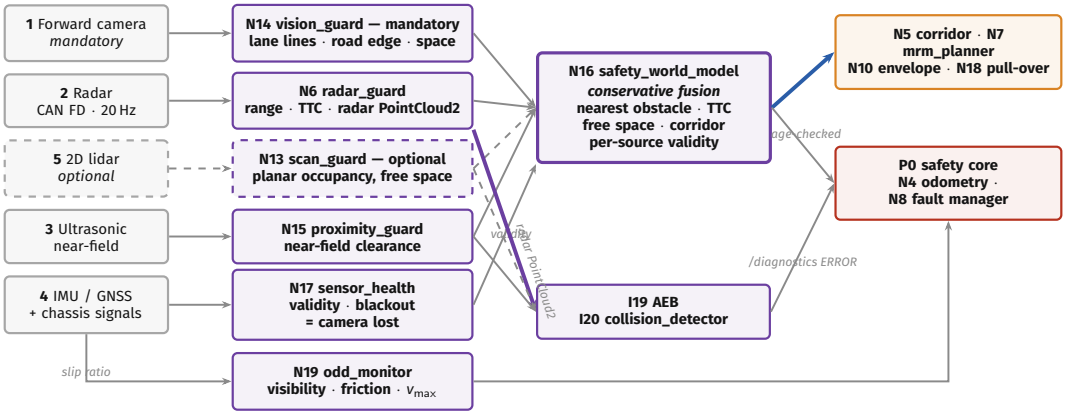
1. **Independent harness and power.** Island sensors must not share a rail, a bundle or a switch with the HPC's set — *a common-cause failure that blinds the HPC must not blind the island.*
2. **One-way data flow.** Sensors feed the island. No island sensing node may ever depend on an HPC-produced topic.

# Five new sensing nodes

| # | Node | Purpose | Rate |
|---|---|---|---|
| N13 | si_scan_guard | *Optional.* 2D lidar ingest. Fixed-size planar occupancy; free-space distance and nearest in-path return. Publishes a **bounded `PointCloud2` for I19/I20**. Classical, no learning, static memory | 10–40 Hz |
| N14 | si_vision_guard | **The one mandatory sensing node.** Lane boundary polylines, free-space edge, road edge and the drivable lateral space — B3's pull-over space; feeds N5 as the required corridor once the island drives | 10–20 Hz |
| N15 | si_proximity_guard | *Rank 3, where fitted.* Ultrasonic ingest. Near-field clearance per sector; gates creep, standstill release, the final metres of a pull-over | 10–20 Hz |
| N16 | si_safety_world_model | **The fusion point.** Combines N6, N13, N14, N15 into one bounded safety world state. *Conservative fusion:* a hazard from **any** source is taken as present; a corridor is trusted only when sources agree and the camera is one of them | 20–50 Hz |
| N17 | si_sensor_health | Per-sensor availability, frame rate, blockage, degradation — dirt, spray, fog, low sun, radar interference. Sets each source's validity flag for N16. Declares blackout = camera lost | 5–10 Hz |

**Takeaway.** Without D7 the island could execute a *blind* MRM. With D7 it executes an **obstacle-aware** one — brake harder for a detected obstacle, stop short of it, keep lane on perceived boundaries after the map corridor expires. Two of the five, N13 and N15, exist only where their optional sensor is fitted.

# The island sensing chain




Solid is the ranked sensors (ultrasonic, rank 3, from rung C); dashed exists only where the optional 2D lidar is fitted. Two paths out of N6: a bounded `PointCloud2` straight to the migrated AEB node — which is what lets it run unmodified — *and* the fused world state for the supervision tier. The safety core reads N16 only through an age-checked snapshot.

Pull-over MRM

The HPC nominates refuges. The island verifies one, executes it, and can abandon it at every step.



# The island cannot decide *where* to pull over

Choosing a legal stopping place means knowing about intersections, crosswalks, bus stops, bridges, tunnels and local traffic law. That is map and planner work, and it stays on the HPC.

**The split**

**The HPC continuously nominates refuges. The island verifies and executes one.**

*… but under* B3 *the HPC is often gone* **new in 20260918** *— next slide.*

**The refuge contract — H8**

| | |
|---|---|
| `side` | LEFT / RIGHT |
| `entry_s, exit_s` | Longitudinal window, as arc length |
| `lateral_offset_m` | Target displacement from lane centre |
| `refuge_width_m` | Usable width, for clearance |
| `surface_class` | PAVED / GRAVEL / UNKNOWN |
| `legality` | LEGAL / EMERGENCY_ONLY / PROHIBITED |
| `validity_distance_m` | How far ahead it remains valid |
| `fallback_behavior` | Normally IN_LANE_STOP |

**The one hard guarantee**

At any moment the buffered refuge set must cover **at least the current worst-case stopping distance** plus the island's fault-reaction budget.

When the HPC cannot honour that — a tunnel, a bridge, a barriered section — it must **say so explicitly** rather than fall silent.

**Takeaway.** Consequence: **the island always knows, *before* any failure, whether a pull-over is available.** N18 maintains a continuous `pull_over_available` flag — fallback capability becomes an *observable*.

# When the HPC offers no refuge: the island finds one

new in 20260918

B3 is reached from B2 exactly when the HPC has been **lost** — and H8's buffered refuge set ages out like every other buffered input. An island with no second source makes the refuge contract a **single point of failure for the one manoeuvre it exists to enable**.

| | H8 nominated refuge | N18 island-detected refuge |
|---|---|---|
| **Physically free** | verified by N16, before use | verified by N16, before use |
| **Legal** | checked against map and traffic law by the HPC | **unknown** — the island holds neither |
| **`legality`** | `LEGAL_STOP` or `EMERGENCY_ONLY`, as nominated | `EMERGENCY_ONLY`, **always** |
| **Admissible** | whenever offered and within `validity_distance_m` | **only** when no nominated refuge is available |
| **Sensors** | N16 free space | N14 **camera** — lane lines, road edge, space between + N6 radar; N15 near-field for a tight-to-kerb end, N13 only to veto, where fitted |

**The asymmetry is the whole argument**

An island-detected refuge is a **physically clear lateral space** and nothing more — a verge, a hard shoulder, a wide kerb, and equally a cycle lane, a bus stop or a driveway, which the island *cannot tell apart*. Admitting one means preferring an unlawful stopping place to a stop in a live running lane.

**Search, not selection**

The island takes the **first** lateral clearance that passes verification, inside $T_{\text{refuge}}$. It weighs no candidates against each other for legality, and defers no stop to reach a better place further on — which is what keeps map and planner semantics off the island (R4, R5).

**Takeaway.** Three outcomes, and they are *not* interchangeable: **found** → pull over; **positively none** → in-lane stop, the corridor still in hand; **inconclusive at** $T_{\text{refuge}}$ → B4, because an inconclusive search is a statement about *capability*.

# Two nodes carry it — and Autoware already has the slot

| # | Node | Purpose | Rate |
|---|---|---|---|
| N18 | `si_pull_over_manager` | Implements Autoware's **existing** `pull_over_manager` MRM slot — `OperateMrm` service server, `MrmBehaviorStatus` publisher — so I7 selects it natively. Owns refuge go/no-go against N16, longitudinal shifting within the window, the availability flag, and the abort ladder | 20 Hz |
| N7 | `si_mrm_planner` *(extended)* | Was a longitudinal ramp. Now emits a bounded `autoware_planning_msgs/Trajectory` ($\leq$50 points) carrying **both** the lateral displacement path and its velocity profile. Hard bounds on lateral acceleration, steering rate and jerk applied here | 50 Hz |

Extending N7 to emit a standard `Trajectory` rather than adding a separate path planner is the cheaper move: I13 `pure_pursuit` then consumes the fallback path through **exactly the same interface, unmodified**, and the fallback tier looks identical to the nominal tier from the gate's point of view.

**Already upstream**

`mrm_handler` defines `/system/mrm/pull_over_manager/status,operate`. **Nothing new is needed for MRM selection.** What *is* missing is the reporting enum: `MrmState.behavior` carries only `NONE`, `EMERGENCY_STOP`, `COMFORTABLE_STOP`.

# The abort ladder

Pull-over must be abandonable at every point into a *strictly simpler* manoeuvre. This is the property that makes it safe to attempt at all.

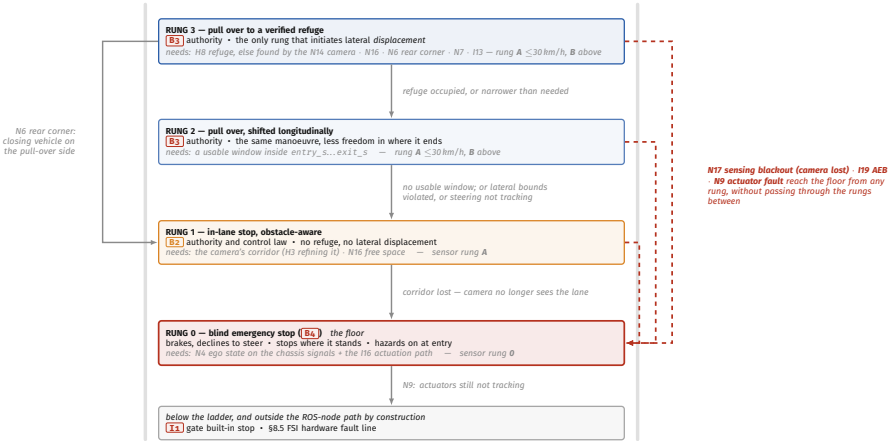
| Trigger | Response |
|---|---|
| Refuge occupied or narrower than needed | Shift within `entry_s…exit_s` |
| No usable window remains | **In-lane stop** |
| Refuge search *concludes*: neither source offers one | **In-lane stop** — the corridor is still there |
| $T_{\text{refuge}}$ elapsed, search still *inconclusive* | **Emergency stop** (B4) |
| Rear-corner radar detects a closing vehicle on the pull-over side | Hold lateral, keep decelerating → **in-lane stop** |
| Lateral path bounds violated, or steering not tracking | Freeze lateral → **in-lane stop** |
| I19 AEB fires | **Emergency stop** (B4) |
| Actuator not responding N9 | **Emergency stop** (B4) |
| Sensing blackout N17 — camera lost: the refuge can no longer be verified free | **Emergency stop** (B4); the in-lane stop is *not* available either |
| Standstill reached, near field not clear | Hold, do not release; lamps stay on — raised at entry |

pull-over → in-lane stop → emergency stop → gate built-in stop

**Takeaway.** Every rung degrades toward something the island can *definitely* do. **The manoeuvre never escalates in complexity under fault.** The two new rungs differ: a search that **concludes** still has a corridor; one that **cannot conclude** is a capability statement, and capability loss goes to the floor.

# The abort ladder, drawn

new in 20260903




**Takeaway.** **Each rung down removes a requirement and never adds one.** That is what makes attempting rung 3 acceptable — the cost of failing at it is bounded by what is underneath, and what is underneath is always strictly simpler.

ODD degradation

Nothing has failed. The world left the envelope — so constrain the HPC before replacing it.



# A different trigger class

| | HPC failure | ODD exit — weather / visibility |
|---|---|---|
| Standard | ISO 26262 — something malfunctioned | **ISO 21448 / SOTIF** — nothing malfunctioned |
| HPC state | Dead, hung, or lying | **Healthy and correct by its own lights** |
| Time constant | Milliseconds | Seconds to minutes |
| First response | Seize authority | **Constrain, and let the HPC act** |
| Who pulls over | The island, because the HPC cannot | **The HPC, because it has the map** |

The last row is the important inversion. When visibility collapses, the *best* outcome is that the HPC executes a planned, comfortable, legal pull-over *while it still can*. The island's job is to **demand that, then verify compliance within a deadline** — and take authority only if the HPC does not act.

**Why the island must detect this independently**

An island that only reads the HPC's self-reported perception confidence cannot detect the case that matters: **the HPC is confidently wrong.** No amount of health monitoring reaches it. The only thing that does is an independent sensing channel that degrades *differently*. *This is the strongest argument for the D7 sensor set — stronger than the fallback argument.*

# Detection — concrete signals, no learning required

| Signal | Sensor (rank) | Indicates |
|---|---|---|
| Contrast / MTF collapse; sky-to-scene luminance; flare area; drop and blur detection | camera (1) | Fog, glare, low sun, dirty or wet lens |
| Lane-detection continuity and confidence | camera (1) | Markings obscured by snow, water film, wear |
| **Radar-vs-camera detection divergence** — radar reports targets the camera cannot confirm at the same range | radar + camera (2) | **Visibility collapse, by physical diversity rather than self-report** |
| **Longitudinal slip ratio vs. commanded acceleration** | wheel speed + IMU (4) | **Low friction — ice, standing water** |
| Yaw-rate vs. steering-angle model residual | IMU + steering (4) | Reduced lateral grip |
| Range collapse, intensity drop, multi-echo rise; radar-vs-lidar divergence | 2D lidar (5, optional) | Fog, spray, particulates — a second instance, where fitted |
| Wiper state, exterior temperature, rain sensor | vehicle CAN | Cheap corroboration |

**Build the radar-versus-camera divergence test first** — it uses the two highest-ranked sensors, so every island from rung B has it. It is a *physical* diversity argument: radar penetrates what optics do not, so persistent disagreement at the same range **is** reduced visibility. Nothing about it depends on a model of the weather.

Friction matters as much as visibility and is usually forgotten. It bounds achievable deceleration → stopping distance → both the speed limit and the refuge window.

# The speed law

$$v_{\max} = \sqrt{2 \cdot a_{\text{brake}} \cdot \left(d_{\text{visible}} - d_{\text{margin}}\right)}$$

| | | |
|---|---|---|
| $d_{\text{visible}}$ | measured sensing range, minimum over the active modalities | N19 |
| $a_{\text{brake}}$ | achievable deceleration, bounded by the friction estimate | N19 |
| $d_{\text{margin}}$ | reaction and actuation margin at the current speed | |

**You may drive only as fast as you can stop within what you can see**

Worked example — $d_{\text{visible}} = 30\,\text{m}$ in fog, $a_{\text{brake}} = 3\,\text{m/s}^2$ on a wet surface, $d_{\text{margin}} = 5\,\text{m}$:

$$v_{\max} = \sqrt{2 \cdot 3 \cdot 25} = 12.2\ \text{m/s} \approx 44\ \text{km/h}$$

**Takeaway.** **Both inputs are measured on the island**, so the limit is defensible without trusting the HPC. Closed-form, auditable, and appropriate for an ASIL-B node — no lookup table.

# The graded degradation ladder

| ODD margin | Island action | Mechanism | HPC deadline |
|---|---|---|---|
| Nominal | none | — | — |
| **Marginal** | Notify. Publish the reduced verdict; HPC re-plans conservatively | `OddVerdict` + mode `ALLOW` | — |
| **Unsatisfiable** | **Demand an ODD switch.** The HPC must move to a domain it can meet. **The island names no target** | `odd_satisfiable = false` | $T_{ack}$ 2 s declare $T_{odd}$ 10 s comply |
| **Degraded** | Restrict. Hard speed cap from the law; enlarge margins — held *underneath* the switch, not instead of it | mode `CONSTRAIN` | 3 s to comply |
| **Beyond ODD, HPC compliant** | Request MRM. Ask the HPC to pull over *while it still can* — and where *no domain is satisfiable* routes | mode `MRM` + refuge request | 10 s to begin60 s to complete |
| **Beyond ODD, non-compliant** | Take authority — island pull-over | `I7` → `N18` | — |

### Compliance supervision

The island issues a constraint, then watches whether actual speed converges within the deadline. **Non-compliance is itself a fault** — and one the island detects *without any HPC cooperation*, from vehicle status it already receives.

### Asymmetric hysteresis

Weather flickers. `N19` degrades on ∼2 s of evidence and recovers only after ∼30 s of clear evidence. **Degrade fast, recover slowly** — and recovery additionally requires that the HPC's own verdict agrees.

# Two nodes, two questions — and where the features meet

| Node | Question | Answer, and where it escalates |
|---|---|---|
| N17 `si_sensor_health` | **"Is my sensor working?"** A *device* question — blockage, dirt, frame rate, internal fault | Per-source validity flag → the **fault manager**. A blinded sensor is a *fault* |
| N19 `si_odd_monitor` | **"Is the world outside my envelope?"** An *environment* question | ODD verdict with a margin → the **degradation ladder**. Reduced visibility is *not* a fault |

### Verdict arbitration

- **The more restrictive verdict binds.** Either domain may restrict; neither may relax the other.
- **Disagreement is itself a signal.** HPC says in-ODD, island says out-of-ODD — precisely the confidently-wrong case. Recorded by N11 as a SOTIF event: evidence about the HPC's perception that no other mechanism produces.

**Fallback availability is an ODD dimension**

If N18 reports `pull_over_available = false` — tunnel, barriered section, no shoulder — **and** N19 reports falling visibility, the vehicle is entering a stretch where the degraded fallback is *in-lane stop on a road it cannot see.* Neither signal alone justifies much; together they justify a much tighter speed ceiling, or not entering at all.

Interfaces and deployment

Ten signals in, nine out,
and not one of them a hard
dependency of the safety core.



# HPC → island: ten typed signals

| Signal | Type | Rate | Age limit | Protection |
|---|---|---|---|---|
| Trajectory (≤250 pts) | `autoware_planning_msgs/Trajectory` | 10 Hz | 300 ms | SI-E2E/A |
| Drivable corridor *(preferred)* | `SafetyCorridor` | 10 Hz | 300 ms | SI-E2E/A |
| Pull-over refuge | `PullOverRefuge` | 2 Hz | 2 s | SI-E2E/A |
| Kinematic state *(reference)* | `nav_msgs/Odometry` | 50 Hz | 60 ms | SI-E2E/A |
| Acceleration *(reference)* | `AccelWithCovarianceStamped` | 50 Hz | 60 ms | SI-E2E/A |
| Operation mode state | `OperationModeState` | 10 Hz | 300 ms | SI-E2E/A |
| Capability vector | `HpcCapability` | 10 Hz | 300 ms | SI-E2E/B |
| ODD status | `OddStatus` | 10 Hz | 300 ms | SI-E2E/B |
| Heartbeat + self-test | `HpcHeartbeat` | 100 Hz | **20 ms** | SI-E2E/B |
| Auxiliary commands | `gear, turn_indicators, hazard_lights` | 10 Hz | 300 ms | SI-E2E/A |

**Every input is degradable.** Kinematic state → N4 dead reckoning. Corridor → N14 camera lanes — and nothing below: no camera corridor is B4. Trajectory → the buffered last-valid, then N7's generated profile. Refuge → in-lane stop. Capability and ODD → N2 and N17.

> **Takeaway.** **No row on this table is a hard dependency of the safety core.**

# Island → HPC, and island → actuators

| Signal | Type | Rate |
|---|---|---|
| Island safety state | `IslandState` — mode, fault code, authority holder | 50 Hz |
| MRM state | `autoware_adapi_v1_msgs/MrmState` | 10 Hz |
| Constraint / veto vector | `SafetyEnvelopeCommand` — accel, curvature, speed limits; ALLOW / CONSTRAIN / MRM | 50 Hz |
| Executed command (echo) | `autoware_control_msgs/Control` | 50 Hz |
| Vehicle status | `velocity, steering, gear, control_mode` | 50–100 Hz |
| Safety world state | `SafetyWorldState` — nearest obstacle, TTC, free space, corridor, validity | 20 Hz |
| Island ODD verdict | `OddVerdict` — visibility, precipitation, friction, derived $v_{max}$ | 2–5 Hz |
| Fallback availability | `FallbackAvailability` — `pull_over_available`, reason, distance | 2 Hz |

### Island → actuator gateway

CAN FD or deterministic Ethernet at **100 Hz**, protection **SI-E2E/B**. Actuator feedback returns on the same bus for N9. **This path shares nothing with the HPC link.**

### On-SoC FSI → island

SPI plus a discrete SOC_ERROR GPIO — hardware faults, watchdog expiry, failover triggers — **bypassing Linux and the TSN switch entirely**. Sub-millisecond detection; the 20 ms heartbeat is the *vehicle-level-independent* path.

# In-house safety-message protection

D6 originally named AUTOSAR E2E. The WG has scoped this design as **Autoware in-house**, so AUTOSAR is out. **The obligation is not:** it comes from the communication threat analysis in EN 50159 and ISO 26262-6, and it does not care which implementation carries it.

### What the receiver must still detect

| Threat | Detected by |
|---|---|
| Corruption | CRC |
| Unintended repetition | counter |
| Incorrect sequence, reordering | counter |
| Loss | counter gap |
| Unacceptable delay, staleness | timestamp vs. age limit |
| Insertion, masquerade | Data ID |
| Addressing error | Data ID |
| Truncation, malformed payload | length |

### Two profiles cover the whole interface

| | SI-E2E/A | SI-E2E/B |
|---|---|---|
| Used for | Ethernet, variable length | compact status, CAN FD |
| Data ID | 16-bit | 8-bit |
| Counter | 8-bit | 4-bit |
| Length | 16-bit | implicit |
| Timestamp | 64-bit gPTP | 32-bit ms |
| CRC | CRC-32 | CRC-8 |
| Overhead | 17 bytes | 6 bytes |

### The rule that actually matters

`N1` verifies on ingress, `H1` applies on egress, and **a verification failure is a safety event raised to `N8` — never a middleware exception.** That is independent of which profile carries it.

nano-ros already lists an E2E protocol with CRC and an EN 50159 mapping. **Part 3's first task is to check whether it covers SI-E2E/A and /B outright** — if so the island inherits it, along with the middleware's own verification evidence, instead of implementing a protocol.

# Execution partitions — priority encodes the degradation strategy

| Partition | Nodes | Period | Priority |
|---|---|---|---|
| P0 Safety core | I1, I2, I3, I5, I6, I7, I16, N1, N2, N3, N4, N8 | 10 ms (100 Hz) | 1 |
| P1 Supervision + fallback | I8, I13, N5, N7, N9, N10, N12, N18 | 20 ms (50 Hz) | 2 |
| P2 Safety sensing | I19, I20, N6, N13, N14, N15, N16, N17, N19 | 20–50 ms | 3 |
| P3 Nominal control | I9, I10, I11, I12, I14, I15 | 20–30 ms | 4 |
| P4 Non-real-time | N11, telemetry egress | 100–1000 ms | 5 |

**Why the QM nominal controller sits *below* ASIL-B sensing Under overload the island sheds the expensive MPC first — the cheap ASIL-B fallback controller in P1 takes over within one cycle, which is exactly what the two-tier design exists to make possible. Shedding sensing instead would silently downgrade an obstacle-aware MRM to a blind one — the worse outcome.**

**Takeaway.** **P0 must complete without P2 or P3.** Already exhibited by the measured failover (9.96 ms mean through a Linux outage); Part 2 must *prove* it rather than observe it.

# Fault-reaction budget, and the nano-ros realisation

## Preliminary budget

| | |
|---|---|
| Heartbeat / deadline miss N2 | $\leq$30 ms |
| FSI hardware fault line | $\leq$1 ms |
| Decision N8 $\rightarrow$ N3 $\rightarrow$ I2/I3 | $\leq$20 ms |
| Command I1 $\rightarrow$ I16 $\rightarrow$ CAN FD | $\leq$20 ms |
| **Island reaction total** | $\leq$**70 ms** |
| Actuator response | vehicle-specific |

The 500 ms watchdog in the evaluation island is a coarse outer bound; 30 ms heartbeat detection is the production target, with the watchdog kept as the independent backstop.

## What nano-ros buys the allocation

- `no_std` **Rust** — D5's "memory safe" at the *language* level
- **148 Kani** BMC harnesses + **83 Verus** proofs over CDR serialization, scheduling, protocol correctness — reusable ISO 26262 evidence for the hardest element
- Cyclone DDS backend **wire-compatible with stock ROS 2 RTPS** — no bridge, no type adaptation
- RMW, domain and topology are **configuration, not code**
- Verified on Zephyr, FreeRTOS, NuttX, ThreadX, bare-metal

**The allocation-policy decision**

XRCE-DDS allocates *only at entity setup*; zenoh-pico needs a bump arena; Cyclone DDS allocates *per message*. `rmw-alloc-sites.py` makes the bound **auditable rather than asserted**.

## Minimal configuration

The smallest island that can still pull the vehicle over: 21 nodes, four sensors.



# Minimal sensor set for a pull-over MRM

A ladder, not a single answer — it depends on the declared ODD.

| Rung | Sensors | Sufficient for |
|---|---|---|
| **0** *the floor* | chassis signals only — or a lost camera, whatever else is fitted | B4 and nothing else: a blind stop in the running lane. **Not a configuration anyone deploys** — it is what A–E degrade *to* |
| **A** irreducible | **forward camera** + chassis signals | B2 at the city $v_{odd}$ ($\sim$30 km/h); B3 at $\lesssim$30 km/h, the camera finding lane lines and space; fair weather. **Where the NTU golf-cart prototype starts** |
| **B** design point | + forward radar, rear-corner radar (pull-over side) | Highway $v_{odd}$ ($\sim$60 km/h), B3 above 30 km/h, weather robustness, closing speed — and the radar-vs-camera test B1 needs |
| **C** near-field | + ultrasonic array | Tight-to-kerb end of B3, not just within the shoulder; standstill / creep release; I20 runs on N15 |
| **D** full | + island IMU / GNSS | Dynamics independent of the chassis bus; friction margin; GNSS corroborates the ODD class |
| **E** enriched | + 2D lidar — *optional* | Second free-space channel; edge confirmation. **Adds confidence, never capability** |

**Takeaway.** **D8 inverts the earlier ladder:** the camera is irreducible and the 2D lidar optional. Every rung above A adds speed, ODD width or a tighter end; rung E adds only confidence.

# Minimal sensor set — why each one

| Sensor | Verdict | Why |
|---|---|---|
| Forward camera | **cannot be dropped** | Every behaviour above B4 is about the *lane*: B2 holds it, B3 finds the space beside it. Only the camera sees lane lines; its loss *is* B4 |
| Chassis signals | **always present** | Wheel speed, steering, ESC yaw / accel: N4's ego state after HPC loss |
| Rear-corner radar | **mandatory >30 km/h** | The manoeuvre crosses lateral traffic space a forward camera cannot see |
| Forward radar | **rank 2; highway + weather** | The only forward channel surviving fog, rain, darkness; AEB's point cloud; half the visibility test |
| Ultrasonic | **rank 3** | The only sensor that sees the kerb or a bollard in the last metre; buys a tight-to-kerb end and standstill release, not speed |
| IMU / GNSS | rank 4, recommended | N4 off the chassis bus; friction margin; ODD-class corroboration |
| 2D lidar | **optional** | Would confirm the camera's refuge well — so it may confirm or veto, never be required |

**Takeaway.** A behaviour written against an optional sensor is not deployable on the islands that lack it. So **only the camera may be required**; everything below it buys speed or ODD width.

# Minimal node set — 21 mandatory

| Group | Nodes | # |
|---|---|---|
| Authority and state | I1 gate · I2 decider · I3 switcher · I7 mrm_handler · N2 supervisor · N3 state machine · N8 fault manager | 7 |
| Ingress, ego state, actuation | N1 gateway · N4 vehicle state estimator · I16 vehicle interface | 3 |
| MRM behaviours | N18 pull_over_manager · I6 emergency stop *(abort)* · I5 stop_mode_operator *(terminal + auto-park)* | 3 |
| Fallback control | N7 mrm_planner *(bounded Trajectory)* · I13 pure_pursuit | 2 |
| Sensing | N14 vision · N6 radar · N16 world model · N17 health · I19 AEB | 5 |
| Execution assurance | N9 actuator supervisor | 1 |

**Recommended (+3):** I9 + I12 + I11. The MRM needs a longitudinal law — reusing Autoware's PID beats writing one, for the same provenance reason the whole design rests on.

**The two-tier argument, concluded**

I10 `mpc_lateral_controller` is **not** in this set. **The pull-over MRM does not need the MPC. 21 mandatory, 24 as recommended — against 37 in the full design.**

# What is deliberately excluded, and the risk accepted

A minimal set is only useful if the omissions are *argued* rather than overlooked.

| Excluded | Why safe to omit for MRM execution | Risk accepted |
|---|---|---|
| I10 MPC lateral | Nominal comfort controller; the MRM tracks a bounded path with pure pursuit | Ride quality. Irrelevant |
| I8 validator, N10 envelope | N7 applies hard lateral-accel, steering-rate and jerk bounds at generation time | **Loss of the guardian role** — can still fall back, cannot veto a *healthy* HPC. Restores in phase 2 |
| N5 corridor monitor | Pull-over deliberately leaves the lane; departure checking must be suppressed anyway | No lateral containment check during nominal driving |
| N12 time quality | After HPC loss the gPTP master may be gone; age checks become local | Cross-domain freshness claims unavailable — and by then there is no cross-domain traffic |
| N11 event recorder | Not needed to *execute* | **Audit (step 4) unmet.** Required before a pilot, not before a bench demo |
| I14 shift decider | I5's `enable_auto_parking` covers gear P | none |
| I15 raw cmd converter | Only for pedal/torque interfaces. **Not needed on the NTU golf cart** (steering angle + velocity) | Vehicle-specific; re-add per platform |
| N13, I20, N15 | Rungs A–B have no ultrasonic (rank 3) or lidar (optional); I19 runs on radar | No near-field guard for the final metres; no second free-space channel. Pair with a "stop within the shoulder" terminal condition |
| N19 ODD monitor | Not required to execute an MRM | The feature is simply absent |

Two design branches

Thirteen questions for the WG,
and the shortest credible path
to a demonstrable fallback.



# Two branches, one design

new in 20261023

Both branches share the six behaviours, the node set and the interface contract. They differ only in what is fitted to the island — and since a behaviour may require only the camera, **both deliver all six**. What changes is the speed, the ODD and the evidence each behaviour can claim.

| | Least-Cost | Maximum-Safety |
|---|---|---|
| Intent | Cheapest island that still delivers all six behaviours | As much sensing as the island processor can carry |
| Forward camera | 1 — low-res mono, 100BASE-T1 Ethernet | 1 — GMSL / CSI-2, higher resolution |
| Radar | **none** | **4** — two front corners, two rear corners |
| Ultrasonic | 5 | 5 |
| IMU / GNSS | 1 | 1 |
| 2D lidar | **1** | none |
| Processor | NXP **S32Z2** / Infineon **AURIX TC4x** / ST **Stellar** | Renesas **R-Car X5H** |
| Sensor rung | A, C, D, E **without B** — off the ladder | D, with corner radar on **both** sides |
| Island ODD | **City road, ∼30 km/h fallback, fair weather** | **Highway (∼60 km/h) and city, weather included** |

**Takeaway.** Placement proposals for both: ultrasonic three across the front, two on the kerb side; “two front radars” read as two corner units (decisions 21(c), 23(a)).

# Why Least-Cost is a city-road branch

new in 20261023

Without radar, forward range comes from the camera and a low-cost 2D lidar — a few tens of metres on dark targets, less in rain.

| Speed | Distance to stop for an obstacle | Low-cost 2D lidar ($\approx$ 20–30 m) | Front radar ($\geq$ 70 m) |
|---|---|---|---|
| 30 km/h | $\approx$ 13 m | **sufficient** | sufficient |
| 60 km/h | $\approx$ 43 m | **insufficient** | sufficient |
| 100 km/h | $\approx$ 110 m | insufficient | sufficient (long-range unit) |

0.5 s from detection to brake, including the §9.2 budget; 4 m/s$^2$ obstacle braking.

- **It caps the vehicle's ODD, not only the island's.** B2 must reach $v_{\text{odd}}$ inside its window: 50→30 km/h takes 2.8 s; 100→30 km/h takes 9.7 s and 176 m — past an 8 s window.
- **Weather sets the same limit.** Camera and lidar lose range together in fog, rain and spray: the island can *detect* lost visibility, not *measure* what is left.

**Takeaway.** A Least-Cost vehicle belongs on roads of $\leq$ **50 km/h, fair weather** — shuttles, campuses, city roads, the NTU golf cart (decision 22).

# Behaviours on each branch — B1, B2, B3 new in 20261023

| | Least-Cost | Maximum-Safety |
|---|---|---|
| B1 | **Narrowed.** Verifies only a city-class, fair-weather target. Visibility: camera contrast + **lidar-vs-camera** — they fail together, so it detects loss but cannot back a degraded domain; an unverifiable declaration (e.g. fog) lets $T_{\text{odd}}$ run out. Friction on the island IMU | **Full.** Two **radar-vs-camera** instances, one per front radar; highway and reduced-visibility domains verifiable. Friction on the island IMU |
| B2 | **City $v_{\text{odd}}$ only** ($\sim$30 km/h). Corridor from the camera, the lidar may narrow it; AEB on N13's lidar. No rear sensor: fixed descent $\leq 2\,\text{m/s}^2$, hazards on. The 2.8 s descent leaves most of 8–15 s to the HPC | **Highway $v_{\text{odd}}$** ($\sim$60 km/h) on front-radar range. **Rear radars watch closing traffic during the descent** (100$\rightarrow$60 km/h: 5.6 s, 124 m) and may pick the rate within a band — never late to $v_{\text{odd}}$; an obstacle ahead overrides. AEB on N6 |
| B3 | **Always** $\leq$ 30 **km/h.** Camera finds the refuge; long indicator dwell and low lateral rate; the **side ultrasonic pair** watches the strip entered and may only disqualify; lidar confirms the edge. Tight-to-kerb end. **Instrumented side only** | **Above 30 km/h.** Rear-corner radar on **both sides**: an H8-nominated refuge on either side is usable; the island's own search stays kerb-side. Front-corner radars check the space ahead |

# Behaviours on each branch — B4, B5, B6 new in 20261023

| | Least-Cost | Maximum-Safety |
|---|---|---|
| B4 | **Unchanged** — the loss of the camera. A surviving lidar keeps I19 armed and may steepen the ramp, nothing else; ultrasonics guard the last metre | **Unchanged.** Surviving radars keep I19 armed and may steepen the ramp. **Rear radars may not soften** B4: braking less for a vehicle behind is not more conservative |
| B5 | Camera + lidar scan + ultrasonic; no radar tracks. **Forward only** — nothing looks back. Video encoded **in the camera** as a second stream N22 forwards; lower $v_{teleop}$ | Camera + **all four radars' tracks** (all-round awareness) + ultrasonic. **Reversing at creep permitted.** Video encoded on the X5H application cluster |
| B6 | Unchanged (rung 0). Front ultrasonics guard forward creep; reversing on the operator's eyes alone | Unchanged (rung 0). Radars and ultrasonics guard both directions |

| Node | Least-Cost | Maximum-Safety |
|---|---|---|
| N6 radar / N13 lidar | N13 only — AEB and confirming channel | N6 only — four radars |
| N19 odd_monitor | contrast + lidar-vs-camera, fair weather | contrast + two radar-vs-camera |
| Minimum set | 21, N13 in place of N6 | 21 as listed |

**Takeaway.** The nine-node B4 path is identical on both branches: it consumes no exteroceptive topic.

# Processors — where the load runs

new in 20261023

Load is dominated by N14: about 80 operations per pixel of classical extraction gives ≈ 0.3 Gop/s at 640×400, 15 Hz (Least-Cost) and ≈ 2 Gop/s at 1.3 MP, 20 Hz (Maximum-Safety). Lidar, radar and ultrasonic ingest are megaops. Estimates — Part 2 measures.

| Least-Cost | Published resources | Where N14 runs | Verdict |
|---|---|---|---|
| **S32Z2** | 8× R52 to 1 GHz + Neon; 25 GFLOPS DSP; 19 MB SRAM | DSP / Neon; well under one core | **recommended** |
| **TC4x** | 6× TriCore 500 MHz, lockstep; PPU (ARC EV71) on TC4Dx | on the **PPU** | with PPU |
| **Stellar** | R52+ at 400–500 MHz; no vector unit found | one core, near saturation | + smart camera |

No CSI-2 receiver found on any of the three: Least-Cost uses an Ethernet camera (decision 21(b)).

| X5H cluster | Hosts |
|---|---|
| R52 lockstep, ASIL D | P0–P2: the whole safety chain, N14 at the higher resolution, N6 for four radars |
| A720AE application | P3 QM MPC under the ASIL-D gate I1 — §4.2's two tiers in silicon; P4; N21, N22 |
| NPU | Unused. A learned detector only under §3.5's conservative-only rule |

**Takeaway.** On the X5H compute stops binding. The limits become the R52 share, CAN FD channels for four radars, and **independence: not the HPC's SoC family**.

# What separates the branches

new in 20261023

**One sensor class.**

Every Least-Cost restriction — city class only, fair weather only, B3 at $\leq 30$ km/h on one side, no reversing under B5 — follows from **having no radar**, not from the microcontroller.

> **The first upgrade to price**
>
> **One forward radar** restores the highway range, weather robustness and the radar-vs-camera visibility test — the cheapest addition that changes the most, if Least-Cost proves too narrow (decision 22).

Decisions and roadmap

Thirteen questions for the WG,
and the shortest credible path
to a demonstrable fallback.



# Open decisions for the WG — 1 of 6

1. **Target or legacy command architecture?** Recommendation: the target set (`control_command_gate` + `command_mode_decider/switcher` + `stop_mode_operator`), pinned to a specific Autoware release — the existing port targets Humble-era interfaces.
2. **Is a QM-rated MPC under an ASIL-D gate acceptable?** Determines whether `autoware_pure_pursuit` is in scope for phase 1.
3. **Which sensors, exactly?** D7 settles the principle, D8 the ranking; each deployment's rung and mounting remain — the camera must see the pull-over side's road edge.
4. **Corridor preference order.** HPC driving: map corridor, cross-checked by the camera. Island driving: camera required, map only refines; heading tube withdrawn — confirm, and the age at which each source expires.
5. **Message set ownership.** Ten new types. Proposal: one `autoware_safety_island_msgs` package contributed upstream, so both domains generate from the same `.msg` sources.
6. **In-house protection profiles.** AUTOSAR is out of scope by WG decision, so `SI-E2E/A` and `SI-E2E/B` are proposed instead. Confirm field widths and CRC polynomials — and first, whether nano-ros's built-in E2E already covers them.
7. **One RMW backend on the island, or two?** A `system.toml` decision, not a code decision.

# Open decisions for the WG — 2 of 6

8. **Classical CV, or a certified smart camera?** *(b) recommended* — it collapses N14 to a bus parser, removes learning-based code from the island entirely, and buys supplier ASIL evidence. It costs BOM and vendor lock-in. If (a), the island's ODD must narrow. The camera is now mandatory and must output road edge and lateral space, not lanes alone.
9. **Do island sensors also feed the HPC?** The design says no — one-way, with an advisory `SafetyWorldState` upward. The alternative quietly creates the reverse dependency the independence argument forbids.
10. **`MrmState` has no pull-over value.** `mrm_handler` already defines a `pull_over_manager` slot, but the AD API enum carries only `NONE`, `EMERGENCY_STOP`, `COMFORTABLE_STOP`. Proposal: contribute `PULL_OVER` upstream rather than overloading `COMFORTABLE_STOP`.
11. **Can the HPC honour the refuge coverage guarantee?** A real constraint on the planner — confirm with the planning maintainers before the interface is fixed.
12. **Which ODD verdict binds?** Proposal: the more restrictive, with disagreement recorded as a SOTIF event. HPC-wins removes the only mechanism that catches a confidently-wrong HPC.
13. **What is the island's own ODD?** Narrower than the vehicle's, and it must be a first-class requirement. Includes B2's $v_{odd}$ table: ∼60 km/h highway, ∼30 km/h city, per sensor rung.

# Open decisions for the WG — 3 of 6: the behaviour set

14. **B2's hold window and hand-back criteria.** $T_{hold}$ — now **8–15 s** (was 2–5 s), since it must contain the descent to $v_{odd}$: 5.6 s from 100 to 60 km/h at 2 m/s$^2$; the value per road class is open — $v_{hold}$, the recovery confirmation window — and the policy question of whether a recovered HPC may be trusted at all within the same drive cycle. **Blocking.**

15. **B4's blackout declaration and stop rate.** *New in version 20260903.* The debounce before N17 declares blackout (now: camera lost) — too short and a tunnel mouth stops the vehicle on a live carriageway, too long and it travels blind — plus $a_{emg}$, the steering-relaxation rate, and whether a blind stop must also *alert* (remote assistance, V2X, horn).

# Open decisions for the WG — 4 of 6 new in 20260918

17. **The ODD switch exchange (B1).** $T_{ack}$ = 2 s and $T_{odd}$ = 10 s are starting proposals only — $T_{odd}$ is additionally bounded by $t_{margin} - t_{pullover}$, so what the WG fixes is a **ceiling and a default**, not one number. **Who owns the ODD catalogue**: the design says the HPC, and the island never names a target — which keeps planning off the island, but means it cannot tell *no satisfiable domain exists* from *the HPC failed to find one*. **What `active_odd`'s bounds must contain**: any bound the island cannot measure is one it cannot verify. And the policy question: **may the vehicle switch domains more than once in a drive cycle?**

18. **The island-detected refuge and $T_{refuge}$ (B3).** The first time the island produces a stopping *place* rather than verifying one. $T_{refuge}$ = 5 s plus a distance-searching cap; the **geometric clearance criteria**, all measured by the camera, since the island cannot classify a surface. The real question: **is an `EMERGENCY_ONLY` clearance acceptable at all?** The island cannot tell a hard shoulder from a cycle lane or a driveway, so admitting it means preferring an *unlawful* stopping place to a stop in a live running lane. The design says yes — second in priority, never displacing a nominated refuge. And **inconclusive versus none** must be implementable: N18 has to be able to say which of the two it reached.

# Open decisions for the WG — 5 of 6: B5, B6 new in 20261023

19. B5 **remote driving and the RCC link.** Stage 1 is designed (§6.6, §8.6). **Ratify the numbers**: $T_{link}$ = 300 ms; command age 600 ms soft, 1.0 s hard; $v_{cap}$ = 20 km/h; a 1.8 s loop; $a_{tele}$ = 3 m/s$^2$. **Re-engaging the ADS** from `SAFE_STOP`: proposed yes, once the HPC is re-qualified and the ODD is satisfiable. **Module and keys**: two SIMs or two modems; who owns the certificate authority; zenoh-pico TLS. **Stage 2**: moving take-over, path-based assistance, RCC human factors.

20. B6 **joystick control.** Stage 1 is designed (§6.6). **Ratify the numbers**: 5 km/h forward, 3 km/h reverse, 0.5 m/s$^2$, $T_{local}$ = 200 ms. **Entry from motion**: never — a moving B5 is stopped first. **The device**: IEC 62745 at PL d, the radio band, the wired port's location — and a **hardwired emergency stop** to the actuator gateway.

# Open decisions for the WG — 6 of 6: the two branches

new in 20261023

21. **Least-Cost hardware (D10). Which microcontroller**: S32Z2 recommended — most headroom for N14, and nano-ros's verified target; TC4x only with the PPU, its integrity argued; Stellar only with a certified smart camera (decision 8). **Camera interface**: no CSI-2 found, so a 100BASE-T1 Ethernet camera — raw $\approx 31$ Mbit/s plus an encoded stream for B5. **Ultrasonic placement** (both branches): three front, two kerb-side.

22. **Least-Cost restricts the vehicle's ODD.** No radar: the island backs $\sim$30 km/h in fair weather and nothing more. Confirm a vehicle ODD of $\leq 50$ **km/h excluding fog and heavy rain** — or price **one forward radar**, which restores highway range, weather and the visibility test.

23. **Maximum-Safety details. Two front radars** read as two corner units. **Either-side pull-over** for an H8-nominated refuge; the island's own search stays kerb-side. **Rear radars in B2's descent**: pick the rate within a band, never late to $v_{\text{odd}}$ — and never soften B4. **Independence**: not the HPC's SoC family; freedom from interference between X5H clusters argued in Part 4.

# Roadmap

| Phase | Adds | Capability reached |
|---|---|---|
| **0** *done* | I9–I12 | Trajectory following on the island, closed loop against unmodified Autoware |
| **1** | P0: I1–I3, I5–I7, I16, N1–N4, N8 | **Autonomous fallback.** Authority transfer on HPC loss; blind but bounded MRM — this is B4 |
| **2** | P1: I8, I13, N5, N7, N9, N10, N12 | **Guardian role.** Command validation, safety envelope, veto, two-tier control |
| **3** | P2: I19, I20, N6, N13–N17 + sensor harness | **Obstacle-aware MRM.** The island stops for what it can see |
| **4** | N18 + H8 | **Pull-over MRM.** The island moves the vehicle out of the running lane |
| **5** | N19 | **ODD degradation.** Independent visibility and friction verdict; graded restriction of a healthy HPC |

**Shortest credible path**

**Phases 1, 3 and 4 reduced to their mandatory nodes deliver a pull-over-capable island in 21 nodes**, deferring the phase-2 guardian role — the sequence recommended for the golf-cart prototype.

**Next parts:** 2 — timing and FTTI, RT contracts. 3 — message definitions, E2E configuration, island state machine. 4 — safety case, HARA, ASIL decomposition, fault-injection matrix.

## Discussion

Full design document, diagrams and node tables: `L4SafetyIsland_SoftwareDesign.md`